

Adithya Venkata Narayanan

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 M.Sc. Robotic Systems Engineering, RWTH Aachen ('26-'28)  German: A1 (RWTH Int'l Academy, completion Mar 2027) | English: C2

Mechatronics engineer with several years of experience building assistive-robotics systems end to end: sensing and **analog front-ends, motor control, embedded firmware, and wireless architecture**. Comfortable across the full electromechanical stack, with a particular focus on control systems, from **adaptive and model-based control (Imperial College London, undergraduate thesis)** to **real-time motor control on STM32 and ESP32**. Seeking Werkstudent or research (HiWi) roles in mechatronics, robotics, and control.

WORK EXPERIENCE

THRYV Mobility Pvt. Ltd. | IIT Madras 

May 2022 – Jul 2026

Mechatronics Team Lead (2024–'26) | Embedded Electronics Intern (2022, '23-'24) under Dr. Manish Anand

Chennai, India

Progressively owned three assistive-mobility products and a shared wireless platform architecture from prototype to production across intern and lead roles, leading embedded teams of 2 to 3 and coordinating with mechanical and electronics teams on requirements, interfaces, and deployment.

Smart-Wheels: *BLE-enabled, hub-integrated power-assist wheel system (BLDC drive, sensor fusion, embedded control)*

- **Led end-to-end development** of the drive platform overseeing the architecture, timelines, milestones, and budget while integrating BLDC drive, load-cell sensing, and embedded control electronics.
- **Designed the closed sensing and actuation loop:** load cells (push-rim/wheel-rim) through an instrumentation-amp and op-amp signal-conditioning chain into a high-res sigma-delta ADC (few-hundred-gram sensitivity), streamed over BLE to the drive unit, which amplifies user torque under safety current clamps and shapes the natural exponential coastdown by actively controlling decay rate to extend glide before rest.
- **Designed the platform's control and sensing circuits** with layout optimized for thermals, compactness, cost, and assembly.
- **Exposed live tuning via the Thryv App** for assist thresholds (min. torque and speed), amplification factor, safety current clamp, and coastdown decay rate.

Three-Way Power-Assist: *Smart drive platform with wireless control and adaptive state logic*

- Led a 2-member embedded subteam to rapidly prototype the drive platform, building a state-machine architecture to manage motor-control logic, mode switching, and peripheral coordination.

Gesture Control Unit: *Wearable gesture interface for assistive mobility systems*

- **Designed the wrist-worn device from schematic to PCB** (ESP32-C3; migrated IMU from ICM-42688P to LIS2DH then LSM6DSx for tap detection and 6-DOF sensing at lower cost and power), evolving it across iterations into a production control interface.
- Diagnosed the power-path charger IC as the dominant quiescent-draw contributor and **designed a custom MOSFET power-gating circuit for true load-disconnect**, optimizing off-state current to nA levels while allowing the device to charge-without-wakeup.
- Designed a press-and-hold sleep/wake feature across the analog power path and MCU; **developed power-optimized BLE firmware (light sleep + bond-parameter tuning)** and optimized the BOM for cost and size across both low-volume R&D and mass manufacturing.
- **Interfaced with the Thryv App to configure tap threshold, stream/log IMU data, and report battery SOC/SOH/TTE/TTF** from an on-board fuel gauge.

Wireless, Security & OTA Platform (ecosystem-wide)

- **Designed the shared BLE control/drive architecture so any control unit (Gesture Control, front console) interfaces interchangeably across all drive platforms**, decoupling control from drive hardware; migrated inter-device comms from ESP-NOW to BLE and moved link-loss detection to BLE supervision timeout and connection parameters, replacing external watchdog/keepalive.
- **Built application-level device authentication** (SHA256-HMAC attestation, eFuse-burned keys, account-gated retrieval) for encrypted, MITM-resistant, mutually-authenticated bonding, working around Android's block on stack-level BLE bond auth, with persistent bonding via OOB exchange and RPA/IRK identity resolution.
- **Developed an Android Thryv App end to end, a tool for the full device fleet:** authenticated BLE bond creation (128-bit OOB), bond management, advertising/scan control, OTA firmware transfer (32-bit CRC, per-session authentication gating), and real-time telemetry.
- Logged phone GPS (latitude, longitude, speed, bearing) alongside wheel RPM and ESP-side IMU orientation into CRC-checked CSVs, giving downstream analysis an independent reference to cross-check device-reported motion against.

Mentored and trained 2 engineering trainees and 6 interns across the Smart-Wheels, Three-Way Power-Assist, and Gesture Control efforts, supporting embedded development and project continuity.

Imperial College London (Remote)

Jun 2023 – Feb 2024

Mechatronics Research Intern under Dr. Rejin John Varghese, Human Robotics Group

Self-Balancing Wheelchair: *Contributed to the simulation and proof-of-concept prototyping.*

- **Deployed a gain-scheduled, model-free adaptive PID controller (MISO) in ROS for tilt and yaw stabilization** under dynamic conditions.
- **Built the simulation-and-validation loop in Gazebo** (physics-based constraints, real-time feedback) **and a data-acquisition node interfacing a VESC and BNO085 IMU** for real-time state feedback.
- **Achieved stable balance up to 52 seconds (yaw within 5°, tilt oscillation 2–5° from setpoint);** traced the cause of instability to incorrect joint definitions in the model.

PROJECTS

Atom (Proof of Concept Development - R2D2 Laboratory)

Apr 2026 – May 2026

Cost-effective, joystick-controlled motorized wheelchair with real-time safety monitoring and speed scaling.

- **Developed the core motor-control firmware on an STM32G030** (ARM Cortex-M0+), leveraging DMA-driven ADC sampling (1 kHz) and TIM1 dual-channel PWM generation to execute a highly responsive arcade drive algorithm.
- **Implemented real-time signal processing**, including exponential smoothing ($\alpha = 0.3$) and deadzone filtering to eliminate input drift, alongside a 3-stage speed configuration managed via **EXTI** button interrupts.

- Engineered a memory-efficient, bufferless SSD1306 OLED driver with a custom 5x7 font to render live system telemetry, while integrating low-voltage and limit-switch interrupt fail-safes to cut motor power during critical faults.

FOC Motor-Controller Research & Design (Thryv + Independent)

Feb 2025 – May 2025

Design-and-research effort spanning production controller design for Thryv and independent FOC study.

- Studied FOC theory in depth, covering Clarke/Park transforms, current-loop structure, and their circuit-level implications, to inform gate-driver and sensing choices; designed and fabricated a DRV8300-based testbed for independent controller development.
- Evaluated integrated FOC driver ICs (TI MCF8329A/MCF series vs. ADI Trinamic TMC9660/TMC6460) on rotor-position estimation and sensed/sensorless tradeoffs to guide Thryv's controller roadmap.
- Drove component and layout decisions for Thryv's BLDC motor controller through comparative study of open-source FOC platforms (VESC 4.12/6.7/MK5, ODrive, Moteus) and reverse-engineering of the FSESC, mapping their design choices to thermal, EMI, compactness, and cost constraints.

4-DOF Lower-Limb Exoskeleton (Undergraduate Thesis)

Jan 2024 – May 2024

Development of a low-cost lower-limb exoskeleton prototype for strength augmentation under Dr. D. Ezhilarasi

- Contributed to the embedded architecture and hardware integration of a 4-DOF lower-limb exoskeleton, performing rigorous system identification on a 1-DOF testbed to model Maxon EC-90 Flat motors and ESCON 50/8 drivers for optimized microcontroller-level deployment.
- Engineered a hybrid closed-loop position controller (Bang-Bang + PD) on an ESP32, leveraging real-time BNO-055 IMU feedback for precise joint coordination while developing active compensators to suppress thigh-induced mechanical disturbances and stabilize inter-joint dynamics.
- Implemented feedforward gait trajectories synchronized with settling-time logic to sustain continuous, fluid gait cycles, validating multi-joint coordinated motion through robust position-tracking speed control (video ).

EDUCATION

Bachelor of Technology, Mechanical Engineering

2020 – 2024 | India

National Institute of Technology, Tiruchirappalli 

CGPA: 8.51/10 | First Class with Distinction

OPEN-SOURCE

VESC-CAN Component on IDF Component Registry

Feb 2026

Authored and published the first VESC-CAN driver on the ESP-IDF Component Registry, targeting the node-based TWAI driver introduced in ESP-IDF v5.5: ISR-safe, fixed-point parsing of the full status-feedback set (integer milliunits, no floating-point) and the complete single-frame command set per Vedder's CAN specification, where existing GitHub implementations target only older IDF versions.

BNO055 Component on IDF Component Registry

Oct 2025

Authored and published the first BNO055 IMU driver on the ESP-IDF Component Registry, a documented, drop-in component with examples, filling a previously unaddressed gap.

NimBLE OTA Example

Sep 2025

Built a BLE OTA update pipeline for ESP32-NimBLE on ESP-IDF, handling packet framing, ring-buffer management, and safe flash-write sequencing for reliable over-the-air firmware updates.

ESP-IDF Hardware Driver Library

Aug 2025

Developed a modular C driver library for sensors, displays, and peripherals on ESP-IDF, with a consistent API surface and structured error handling across components.

ACHIEVEMENTS

GRE General Test '25 (Graduate Record Examinations)

Oct 2025

Educational Testing Service (ETS)

Scored 325 (Q: 170/170, V: 155/170, AWA: 4.5) in GRE General Test (2025)

GATE '24 (Graduate Aptitude Test in Engineering)

Mar 2024

Department of Higher Education & Ministry of Education, Government of India

Achieved 92.7 percentile in GATE 2024 (Instrumentation Engineering), among ~9,900 candidates nationwide.

National Winner of Smart India Hackathon '22

Aug 2022

Ministry of Education, Government of India

Secured the first place, from 2000+ participating teams, in a national-level hackathon organised by the Ministry of Human Resource & Development Wing of the Government of India.

Pre-finalists of e-Yantra Robotics Competition '21

Mar 2022

e-Yantra Labs, IIT Bombay

Made it to the semi-final round with 12 other teams in a national-level robotics competition from 500+ participating teams.

SKILLS

Programming Languages

Embedded C/C++, Python, Lua, MATLAB, HTML, CSS, Kotlin

Frameworks

ROS, ESP-IDF, STM32CubeIDE, Git, Linux, Arduino IDE, Android Studio

Communication Protocols

CAN, I2C, UART, SPI, Bluetooth LE

Design Software

Onshape, EasyEDA, Eagle, Fusion 360, Inventor

Simulation Environments

Gazebo, CircuitLab, CoppeliaSim

Python Libraries

NumPy, OpenCV, Esptool, Mediapipe, Matplotlib, PySerial, Bleak, asyncio